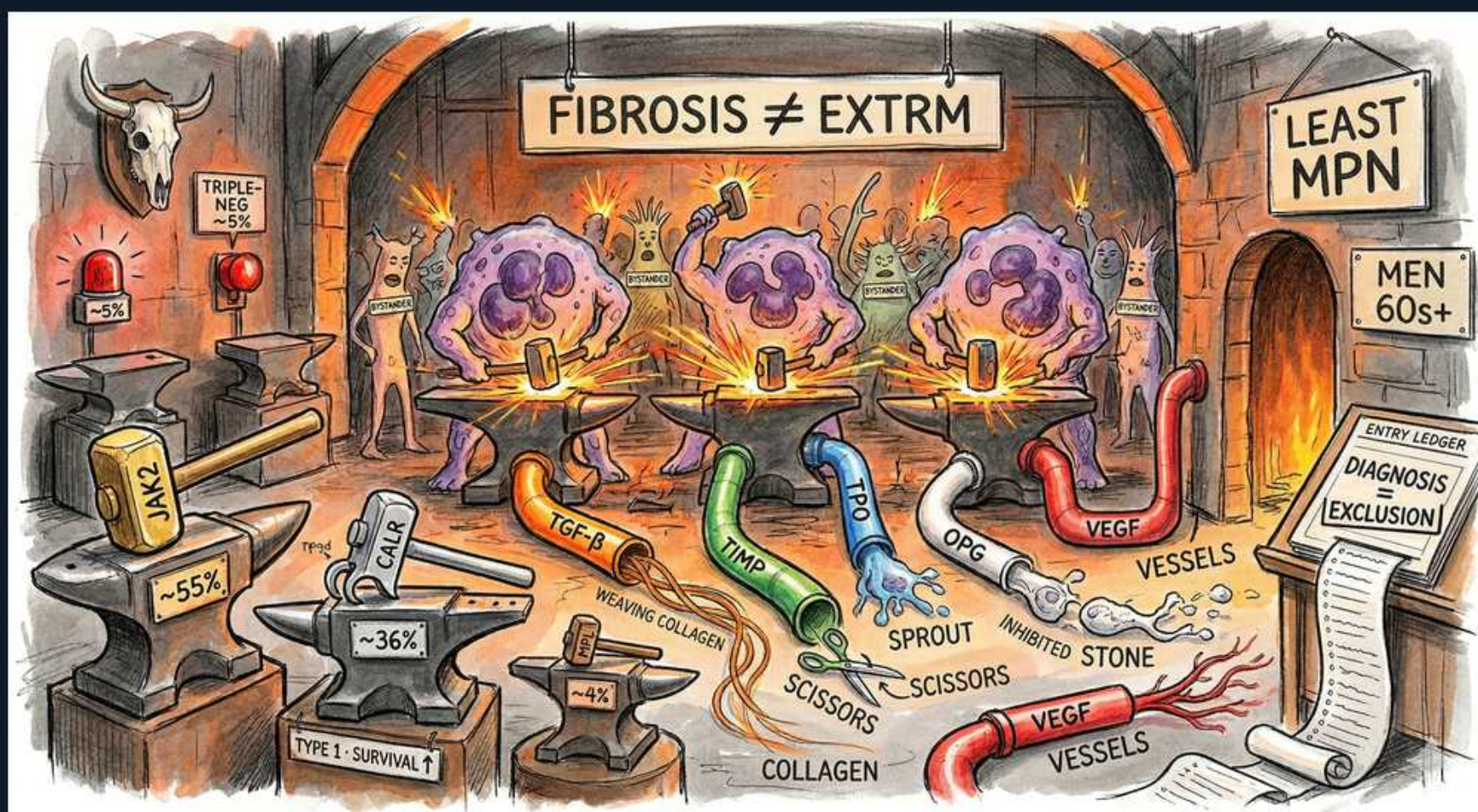


Polycythemia / MPN — Primary myelofibrosis pathophysiology



1. Clonal megakaryocytes

WHAT YOU SEE

Three angry purple muscular megakaryocytes hammering anvils

WHAT IT ENCODES

Primary myelofibrosis (PMF) is driven by a clonal proliferation of atypical, dysplastic megakaryocytes that form dense clusters in the marrow. These abnormal megakaryocytes secrete a storm of fibrogenic cytokines (PDGF, TGF-beta, VEGF, bFGF) that recruit and activate normal fibroblasts. The malignant clone is the megakaryocyte/myeloid lineage — the resulting fibrosis is a downstream reactive consequence, not the cancer itself. Same driver mutations as the other MPNs: JAK2, CALR, or MPL.

BOARD TRAP

WRONG: the fibroblasts depositing collagen are the neoplastic clone. They are NOT — fibroblasts are polyclonal/reactive bystanders responding to megakaryocyte-derived cytokines. The discriminator: clonality markers (JAK2/CALR/MPL, cytogenetics) localize to the hematopoietic megakaryocyte/myeloid lineage, never to the stromal fibroblasts.

2. Fibrosis is reactive, not clonal

WHAT YOU SEE

FIBROSIS ≠ EXTRM banner top

WHAT IT ENCODES

Marrow fibrosis in PMF is a SECONDARY, reactive process. Cytokines released by the clonal megakaryocytes (chiefly TGF-beta and PDGF) drive non-neoplastic fibroblasts to lay down reticulin then collagen. This is the conceptual key to the whole disease: the cancer is hematopoietic, the scar is reactive stroma.

BOARD TRAP

WRONG: fibroblasts are the malignant clone. Discriminator — if you transplant the patient's hematopoietic stem cells into a new marrow, fibrosis recurs from the donor-derived clone driving host fibroblasts; the fibroblasts themselves carry no driver mutation.

3. TGF-beta drives fibrosis

WHAT YOU SEE

Orange TGF-B hose weaving collagen

WHAT IT ENCODES

TGF-beta (transforming growth factor-beta), released by clonal megakaryocytes and platelets, is the dominant pro-fibrotic cytokine in PMF. It stimulates fibroblasts to synthesize type I/III collagen and reticulin and upregulates TIMPs that block collagen breakdown. Mnemonic: TGF-beta = 'Tissue Gets Fibrotic.'

BOARD TRAP

WRONG: TGF-beta is immunostimulatory and promotes marrow regeneration here. In PMF its net effect is profibrotic/immunosuppressive — it shifts the matrix balance toward deposition, not repair.

4. Collagen deposition

WHAT YOU SEE

COLLAGEN rope/threads on floor

WHAT IT ENCODES

Progressive reticulin and then collagen deposition replaces functioning marrow, producing the classic 'dry tap' on aspiration and forcing hematopoiesis out to the spleen and liver (extramedullary hematopoiesis). Fibrosis is graded MF-0 through MF-3 on biopsy; collagen/osteosclerosis marks advanced disease.

BOARD TRAP

WRONG: collagen fibrosis is fully reversible with cytoreduction. Discriminator — ruxolitinib and other JAK inhibitors improve spleen size and symptoms but do NOT reliably reverse established collagen fibrosis; only allogeneic transplant can clear it.

5. TIMP / collagen balance

WHAT YOU SEE

Green TIMP hose, SCISSORS cutting

WHAT IT ENCODES

Tissue inhibitors of metalloproteinases (TIMPs) block matrix metalloproteinases (MMPs), the enzymes that normally degrade collagen. In PMF the TIMP:MMP ratio is pushed toward inhibition, so collagen breakdown falls and net matrix accumulates. Think of TIMPs as taking the scissors away from collagen.

BOARD TRAP

WRONG: high MMP activity is what causes the fibrosis. The opposite — it is MMP INHIBITION (excess TIMP) that tips the balance toward collagen accumulation.

6. PDGF / sprout

WHAT YOU SEE

Blue PDGF hose with SPROUT

WHAT IT ENCODES

Platelet-derived growth factor (PDGF), released by the abnormal megakaryocytes/platelets, is a potent fibroblast mitogen that drives fibroblast proliferation and collagen output, working alongside TGF-beta. PDGF = the 'sprout' growing new fibroblasts.

BOARD TRAP

WRONG: PDGF acts directly on hematopoietic stem cells to cause fibrosis. It does not — PDGF targets the stromal FIBROBLASTS (proliferation/collagen), part of the reactive, non-clonal response.

7. VEGF → vessels (angiogenesis)

WHAT YOU SEE

Red VEGF hose to VESSELS

WHAT IT ENCODES

Vascular endothelial growth factor (VEGF) drives marrow neovascularization (increased microvessel density), a recognized feature of PMF marrow alongside fibrosis. Angiogenesis supports the expanding clone and is part of the disordered marrow microenvironment.

BOARD TRAP

WRONG: increased marrow angiogenesis argues against MPN and points to a benign reactive marrow. On the contrary, marked microvessel density is characteristic of PMF and other MPNs.

8. OPG / osteosclerosis

WHAT YOU SEE

OPG INHIBITED STONE

WHAT IT ENCODES

Cytokine imbalance (including osteoprotegerin/OPG dysregulation favoring osteoblast activity) produces osteosclerosis — thickened, sclerotic bone seen on imaging in advanced PMF. Radiographs may show diffusely increased bone density, especially axial skeleton.

BOARD TRAP

WRONG: PMF causes lytic (punched-out) bone lesions. Those are myeloma — PMF classically causes osteoSCLEROSIS (dense bone), the discriminator on plain films.

9. JAK2 ~55%

WHAT YOU SEE

JAK2 hammer/anvil, ~55%

WHAT IT ENCODES

JAK2 V617F is the most common driver in PMF, present in roughly 50-60% of cases. It is a gain-of-function exon-14 mutation causing constitutive, ligand-independent JAK-STAT signaling. The SAME mutation drives PV (~95%) and ET (~50-60%); JAK2 is shared across all three classic BCR-ABL-negative MPNs.

BOARD TRAP

WRONG: detecting JAK2 V617F confirms polycythemia vera. JAK2 V617F is shared by PV, ET, and PMF — it identifies an MPN but does NOT specify which one; the phenotype (Hct, smear, marrow biopsy) makes the distinction.

10. CALR ~36% (better prognosis)

WHAT YOU SEE

CALR anvil ~36%, TYPE 1 SURVIVAL up

WHAT IT ENCODES

Calreticulin (CALR) mutations occur in roughly 25-35% of PMF and define a JAK2-wild-type subset. Type 1 (52-bp deletion) CALR carries the BEST prognosis of all PMF genotypes — longer survival, lower leukemic transformation. CALR mutations also occur in ET but never in PV.

BOARD TRAP

WRONG: CALR-mutated PMF has a worse prognosis than triple-negative disease. Reversed — CALR (especially type 1) confers the BEST survival; triple-negative PMF has the WORST prognosis.

11. MPL ~4%

WHAT YOU SEE

TYPE 1 ~4% anvil

WHAT IT ENCODES

MPL (thrombopoietin receptor) mutations, typically MPL W515L/K, are found in a small minority (~4-8%) of PMF and ET. They activate the same JAK-STAT axis downstream of the TPO receptor, mimicking ligand binding.

BOARD TRAP

WRONG: MPL mutations are found in polycythemia vera. MPL drives JAK2-negative ET and PMF, not PV — PV is essentially always JAK2-mutated (V617F or exon-12).

12. Triple-negative ~5%

WHAT YOU SEE

TRIPLE-NEG ~5% tag top left

WHAT IT ENCODES

Triple-negative PMF (no detectable JAK2, CALR, or MPL mutation, ~5-10%) carries the WORST prognosis — highest rates of leukemic transformation and shortest survival. These patients often harbor high-molecular-risk subclonal mutations (e.g., ASXL1, SRSF2).

BOARD TRAP

WRONG: triple-negative status means a benign, non-clonal reactive process. Triple-negative PMF is still clonal (driven by other lesions) and is the most aggressive subgroup.

13. Least common MPN, older men

WHAT YOU SEE

LEAST MPN, MEN 60s+ tags right

WHAT IT ENCODES

PMF is the LEAST common of the three classic Ph-negative MPNs (PV and ET are more frequent). It typically presents in the 6th-7th decade with a slight male predominance. Median age at diagnosis is ~65; pediatric/young-adult PMF is rare.

BOARD TRAP

WRONG: PMF is the most common MPN and presents in young women. It is the LEAST common classic MPN and favors older men — useful demographic discriminator on exam vignettes.

14. Diagnosis of exclusion

WHAT YOU SEE

ENTRY LEDGER DIAGNOSIS BY EXCLUSION scroll

WHAT IT ENCODES

PRIMARY MF is diagnosed only after excluding secondary causes of marrow fibrosis (metastatic carcinoma, infection like TB, autoimmune myelofibrosis, hairy cell leukemia, MDS) AND post-PV / post-ET myelofibrosis. WHO criteria require megakaryocytic atypia with reticulin/collagen fibrosis plus a clonal marker (JAK2/CALR/MPL) or another clonal/exclusion criterion.

BOARD TRAP

WRONG: teardrop cells plus splenomegaly are enough to call primary myelofibrosis. Those are non-specific — leukoerythroblastosis and dacrocytes occur in marrow infiltration of any cause; you must exclude secondary fibrosis and prior PV/ET first.